

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 205 — Past Questions & Materials
by Renance

Course: GET 205
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200, 400

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Paper 1 of 2 — GET 205 Fundamentals of Fluid Mechanics PQ

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2017/2018 Academic Session

COURSE CODE AND TITLE: MEE 215 — FLUID MECHANICS |

DATE: JUNE 1, 2018. TIME ALLOWED: 3 HOURS

INSTRUCTIONS: ANSWER QUESTION 1 AND ANY THREE (3) QUESTIONS.

& QUESTION 1

(a)

No doubt, Africa lags other continents in the manufacture of fluid (mechanics) machines. (i) list the reason(s) for this lag (ii)

briefly proffer solutions to the problem identified in a({}) [6 Marks]

'A wooden block of specific gravity 0.7 and having a size of 2m X 0.5m X0.25 m is floating in water. Determine the volume of

(b)

concrete of specific weight 25KN/m³, that may be placed which will immerse the (i) block completely in water and (ii) block

and concrete completely in water. [6 Marks]

(c) Determine the dimension of kinematic viscosity [4 Marks]

(d) A ship to accommodate about 1,006 passengers and fluid machines was designed by the Centre for Maritime and Offshore

Studies, FUPRE. The ship successfully sailed across the Atlantic Ocean. Why did this ship not sink? [1.5 marks]

QUESTION 2

(Q) $w = 2x^? + 4y^3 \sim 27\%$ $4 + 3xt$ models the flow in a crude oil carrying pipeline. Give a complete characterization of the

flow.[4.5 Marks]

(b) 'A fluid flowing through a pipe has an inlet and outlet. The pressure at the inlet is 978 hPa, while at the outlet is 1001 hPa.

Characterize this flow. [4 Marks] *

(c) List four uses of dimensional analysis [2 Marks]

(d) _ Derive an expression for the rate of flow in a venturimeter [5 marks]

(e) __ List two limitations of Dimensional Analysis [2 mark] LOW. pup relolos CHV

QUESTION 3 : C

(a)

(b)

(c)

(d)

(e)

List the applications of dimensional homogeneity? [3 mark]

Which measuring device(s) measure(s) both static and stagnation pressure [3 marks]

List five dimensionless numbers in engineering. [2.5 marks]

Heat is added to a system by change in absolute temperature, thermodynamic relation that defines this process. [4.5 marks]

Characterize the flow defined by this model $u = f(x, y, z, t)$ [4.5 marks]

T leading to a change in entropy, S. Give, Mathematically, the

&Y QUESTION 4

(a)

(b)

(c)

(d)

(e)

% QUESTION 5

(a)

(b)

(c)

(a)

(e)

AY questi

(a)

(b)

(c)

(a)

(e)

What are the three basic ways to solve fluid flow problems? {2 marks]

Define (i) control volume and (ii) system [2 marks]

State, mathematically, the law of conservation of mass [2 marks]

Is Newton's second law a vector or scalar quantity? If yes (or No) state the scalar variants in 1-D, 2-D & 3-D [5 marks]

List the three types of similitude [1.5 marks]

If the surroundings to a system exert a net force, equation. [4 marks]

M about the centre of mass of the system, give the angular momentum

'What is incompressible viscous flow? [1.5 marks]

What does the statement: "a system has a fixed quantity of mass" mean? [5 marks]

Mathematically define unsteady flow. [2 mark]

Mathematically define non-uniform flow? [1.5 marks]

A. U-tube manometer is used to measure the pressure of oil of specific gravity = 13.6 in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the oil in the pipe. [7.5 marks]

ONG

What is Reynold's number? [2 marks]

What is the difference between fluid statics and fluid dynamics? [2 marks]

List five fluid properties. [1.5 marks]

Completely characterize the velocity field: $u = x^2 + 2x + 4$ [5 marks]

Determine the dimension of Gas constant [3.5 marks]

Determine the dimension of shear stress [3.5 marks]

Specific gravity 0.85 flowing in a pipeline. Its left end is connected to the pipe is 100 mm below the level of mercury (specific gravity 13.6). Determine the absolute pressure of the oil in the pipe.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2016/2017 Academic Session

COURSE CODE AND TITLE: MEE 215 - FLUID MECHANICS I

DATE: APRIL 10, 2017. TIME ALLOWED: 2 HOURS

INSTRUCTIONS: ANSWER ANY FOUR (4) QUESTIONS,

QUESTION 1

() Differentiate between Newtonian and non-Newtonian fluids, using a simple sketch only, [2 Marks]

(b) Given that 10 m³ of oil has a mass of 1100 kg. Calculate the following properties of the oil: (i) density, (ii) specific weight, (iii) specific gravity and (iv) specific volume. [6 Marks]

(c) A small cylindrical oil vessel, 1m in diameter, is submerged vertically below the water surface. The top of the vessel is 2m below the water surface. Determine the magnitude of the total resultant force acting on the vessel, and the location of the resultant force. Take specific weight of sea water as 9.81 kN/m³, [9.5 Marks]

QUESTION 2

(a) What is fluid pressure?

(b) Classify the following petroleum products as compressible or incompressible: (i) Kerosene (DPK) (ii) Natural Gas (iii) Acetylene (iv) Premium Fuel Oil (LPFO) and (v) Syngas. Justify your answer,

(c) A U-tube manometer (Fig Q2c) measures the pressure difference between two points A and B in a liquid of density ρ_l . The U-tube contains mercury of density ρ_m . Determine the difference of pressure if $a = 1.5\text{m}$, $b = 0.75\text{m}$ and $h = 0.2\text{m}$. If the liquid in the pipe is water and $p_2 = 13.691\text{ kPa}$, [12 Marks]

pone

sea water

magnitude

centre of

{1.5 Marks]

mpressible fluids: (i) Dual

ium Motor Spirit (PMS) (v)

[4 Marks]

ice between two points A

density p_2 , Calculate the

0.5 m, given that the liquid at A and

Density p ;

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Fig. Qte

QUESTION 3

-(a) The Centre for Maritime and Off-Shore Studies, FUPRE designed a floating vessel. As an Engineer-in-Training, how do you test if this vessel (or body) is stable? [2 mark]

(b) Mathematically characterize the boundary of a system [2 marks]

(c) What is Viscosity? [2 marks]

(d) In the mini Refinery designed and constructed by FUPRE, the rate of flow of the crude oil was not known, To determine the rate of flow, it was suggested that a venturimeter be fitted ak

to the horizontal pipe through which the crude oil would flow, Derive the e:

Xpression for the

rate of flow of the crude oil. Neglect all losses, [9.5 marks] s

QUESTION 4 1

(a) What are the applications of Fluid Dynamics? [3 marks]

(b) Differentiate between a control volume and a control mass [2.5 marks]

(c) Mention two dimensionless numbers in engineering [1 mark]

(d) A group of FUPRE research students solving crude oil flow problems for Dangote Refinery, had suggested that a pipe, called mouthpiece, be fitted to an orifice. The purpose of the mouthpiece, as was suggested by these innovative Engineers-in-training, was to increase the amount of discharge, Estimate the orifice diameter if the length of pipe extension is 400mm. [6 marks]

(e) The velocity of a fluid flow was defined as: $u = \text{beavte}$, Characterize the flow [5 marks]

QUESTION 5 : a)

(a) What is the difference between a com

(b) Show that Bernoulli's equation is dimensionally homogenous? [5 marks]

- (c) State the Buckingham Pi theorem. [2 mark]
- {d} What is dimensional analysis? [1.5 marks]
- (e) Crude oil flows in a spherical pipe. At one section of the Pipe, the diameter is 4m, the static pressure is 265kPa gauge, the velocity is 3.8m/s and the elevation is 10m above the ground level. The elevation at a section downstream is 0 m, and the pipe diameter is 0.15m. Find the gauge pressure at the downstream section, Neglect frictional effects, Assume the density of crude oil to be 1000kg/m³. [7.5 marks]

QUESTION 6

- > (a) What is the difference between a weir and a notch? {1 mark]
- (b) Write short notes on: (i) Incompressible viscous flow (ii) venturimeter meter (v) the continuity equation, [5 marks]
- (c) With the aid of a well labeled diagram, show the shear stress across a section of a pipe carrying a fluid, [1.5 marks]
- (d) Derive an expression for discharge over a rectangular weir [4 marks]
- (e) A horizontal water pipe with diameter 300mm tapers to 150mm diameter, The pressure and mean velocity at the inlet to the tapering section are 50kN/m² and 1m/s respectively. Find the mean velocity and pressure at the outlet of the tapering section, neglecting all losses.

[6 marks]

- (iii) nozzle (iv) orifice

stress and velocity distribution curve

compressible flow and an incompressible flow? [1.5 marks]

: COLLEGE OF ENGINEERING AND TECHNOLOGY

SUEDEPARTMENT OF MECHANICAL ENGINEERING

FIRST SEMESTER EXAMINATION 2022/2023 SESSION

COURSE CODE AND TITLE: MEE 215 BASIC FLUID MECHANICS

TIME ALLOTTED: THREE (3) HOURS

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS

UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

QUESTION 1

- (a) Prove Pascal's law
- (b) Write an expression for total rate of change for compressible fluid
- (c) Write an expression for acceleration component for the compressible fluid
- (d) The diameter of a Pipe at its two sections, section 1 and section 2 are 200 mm and 300 mm, respectively, if the velocity of water through the pipe at section 1 is 4 m/s, find the discharge through the pipe at section 2

[17.5 Marks]

QUESTION 2

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Figure 1: acting forces

on on immersed body

Figure 2 Figure 3

(a) Figure 1 shows forces actin

forces in the x-direction

(b) Write the expression for the acting forces in the z-direction using figure 1

(c) Write the expression for the buoyant forces or Archimede's forces usi 9 figure 1

(d) Calculate specific weight, specific mass, specific volume and specific gravity of a liquid having a volume of 6m³ and weight of 44N,

ig an immersed body. Write the expression for the acting

(17.5 Marks]

QUESTION 3

(a) What are the conditions for the body (see figure 2 above) to float?

(b) What happens if S and D are not on the same vertical line (see figure 3 above)

(c) What is the condition for the body to sink downwards

(d) Given five (5) examples each of fundamental and derived units

(e) A plate 0.05 mm distant from a fixed plate at 1,2 m/s and requires a force of 2.2 N/m?

to

maintain this speed, Find the viscosity of the fluid between the plates.

(17,5 Marks)

QUESTION 4 ches

Figures 4. 5 and 6 shows pressure force on curved surface. Using

figures:

' 6

figure 5 figure

(0) write cn expression for pressure along the x-axis

(b) write on expression for pressure along the y-axis

(c) write an expression for pressure along the z-axis

(d) Define uniform flow in terms of convective acceleration s

(e) Tf the surface tension at air-water interface is 0.069 N/m, what is the pressure difference between inside and outside of an air bubble of diameter 0,009 mm?

(17.5 Marks},

QUESTION 5

(0) List the two approaches to flow field defining and differentiate between the two

approaches

(b) Write the expression for velocity vector and distance in 3-D (Descartes) coordinate System

(c) Define Buoyant force

(d) A single column U-

s

tube manometer made of glass tubing having a nominal inside diameter of 24 mm, has been used to measure Pressure in a pipe or vessel containing air. If the limb opened to atmosphere is 10% oversized. Find the error in mm of mercury in the measurement of air pressure due to surface tension effects. It is stated that merc

fluid for which surface tension $\sigma =$

$4 \times 10^{-3} \text{ N/m}$,

where θ is the contact angle;

$\sigma = 0.52 \text{ N/m}$ and angle of contact $\theta = 140^\circ$ hint: $k =$

(17.5 Marks)

QUESTION 6

(a) Show that along a stream line, stream function is constant

(b) The velocity change in the vicinity of a point can be expressed in terms of partial derivative of the four independent variables (x, y, z, t). Write the expression for (i) the rate of change of velocity in the x, y and z direction, (ii) velocity gradients (iii) convective change

(c) Define and state mathematically the expression for the following: (i) density (ii) stream function (iv) pathlines (v) streamline

(4) In a fluid, the velocity field is given by $V = (3x^2 + 2y^2) \mathbf{i} + (2xy + 4z^2) \mathbf{j} + (2xz + 4y^2) \mathbf{k}$

Determine the velocity component at (1, 2, 3) m/s.

(ii) stream

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FEDERAL UNIVERSITY OF PETROLEUM RESOU EFPuRuy

COLLEGE OF ENGINEERING AND
 FAnSy SEMESTER ExAMENATION SSNS
 "ER EXAMINATION
 COURSE CODE AND TITLE: MEE 215 - paste eLte Asan
 TIME ALLOTTED: THREE (3HR) HOURS
 INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS

QUESTION 1

- (a) Prove Pascla's law
- (b) Write an expression for total rate of change for compressible fluid
- (c) Write an expression for acceleration component for the compressible fluid
- (d) The diameter of a pipe at its Two sections, section 1 and section 2 are 200 mm and 300 mm, respectively. if the velocity of water through the pipe at section 1 is 4 m/s, find the discharge through the @¥ section B
 (17.5 Marks]

QUESTION 2

Figure 3

Figure 1: acting forces Figure 2
 onan immersed body

- (a) Figure 1 shows forces acting on immsrses say Write the expression for the acting forces in the x-direction
- (b) Write the expression for the acting forces in the z-direction using figure 1
- (c) Write the expression for the buoyant forces or Archimede's forces using figure 1
- (d) Calculate specific weight, specific mass, specific yolume and specific gravity of a liquid having a volume of 6m* and weight of 44N.
 {17.5 Marks]

QUESTION 3

- (a) What are the conditions for the body (see figure 2 above) to float?
- (b) What happens if S and D are not on the same vertical line (see figure 3 above)
- (c) What is the condition for the body to sink downwards
- (d) Given five (5) examples each of fundamental and derived units
- (e) A plate 0.05 mm distant from a fixed plate at 1.2 m/s and requires a force of 2.2 N/m +t maintain this speed. Find the viscosity of the fluid between the plates.

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QUESTION 4 |

Fi 4 5nd 6 shows pressure force on curved surface. Using these figures
figures 4, ;

figure 6

Figure 4 figure 5

- (a) write an expression for pressure along the x-axis
- (b) write an expression for pressure along the ywrexis
- (c) write an expression for pressure along the z-axis
- (d) Define uniform flow in terms of convective acceleration
- (e) If the surface tension at air-water interface is 0.069 N/m, what is the pressure difference between inside and outside of an air bubble of diameter 0.009 mm?

(17.5 Marks]

QUESTION 5

- (a) List the two approaches to flow field defining and differentiate between the two approaches.
- (b) Write the expression for velocity vector and distance in 3-D (Descartes) coordinate svstem
- (c) Define Buos ant force
- (d) A single column U-tube mane seter mode of glass tubing having a nominal inside diameter of 2.4 mm, has been used to me sure pressure ine pipe or vessel containing air . if the limb opened to atmosphere is 10% ov. esize Find the error in mmo
f mercury in the measurement
of air pressure due to surface + asion effects It is stated that mercury is the manometric fluid for which surface tension $\sigma = 0.52 \text{ N/m}$ and angle of contact $\theta = 140^\circ$ hint: $A = \frac{40 \cos \theta}{\omega d}$,

QUESTION 6 (17.5 Marks]

- (c) Define and state mathematically he expression for the fol
 - tube (iii) stream function (iv) Pathilines (v) streamline lowing: (i) density (ii) stream
 - (d) Ina fluid. the velocity field is given by $V = (3x^4 - 2y)i + (2x^2y + 3y^2)j + (2x^2 - 3y^2)k$. Determine
- (17.5 Marks}

MEE 215 Problem Set 1 – Due for Submission 3 pm on Thursday, 22/02/2024

1a. Calculate the specific weight w , the density ρ and the specific gravity for oil of which 10 m³ weighs 78.33 KN. Assume g , the acceleration due to gravity is 9.81 m/s², and the specific weight of water is 9.79 KN/m³.

1b. A liquid has a mass density of 1550 kg/m³. Calculate its specific weight, specific gravity, and specific volume.

- 2a. A plate (2m x 2m), 0.25 mm distant apart from a fixed plate, moves at 40 cm/s and requires a force of 1 N. Determine the dynamic viscosity of the fluid in between the plates.
- 2b. At a certain point in an oil the shear stress is 0.2 N/m² and the velocity gradient is 0.21 s⁻¹. If the mass density of the oil is 950 kg/m³ find the kinematic viscosity.
- 3a. Determine the absolute pressure of oil at the point in its storage vessel when gauge pressure reads 80 KPa.
- 3b. Convert a pressure head of 10 m of water column to kerosene of specific gravity 0.8 and carbon-tetra-chloride of specific gravity of 1.62.
- 3c. Determine (a) the gauge pressure and (b) The absolute pressure of water at a depth of 9 m from the surface.
- 3d. A U-tube manometer (illustrated below) measures the pressure difference between two points A and B in a liquid of density ρ_1 . The U-tube contains mercury of density ρ_2 . Given that $a = 1.5$ m, $b = 0.75$ m and $h = 0.5$ m, solve for the difference of pressure in pipes A and B if they both contain water, and given that the specific gravity of mercury is 13.6.
- 4a. An Isosceles triangular plate of base 5 m and altitude 5 m is immersed vertically in an oil of specific gravity 0.8. If the centre of gravity of the triangle is 2.67 m below the free water surface, determine the total pressure and the centre of pressure. Take specific weight of water as 9.79 KN/m³.
- 4b. A wooden block of width 2.5 m and length 6 m is floating horizontally in water. If the wooden block is 0.97 m deep into the water, find (i) the volume of water displaced, (ii) position of centre of buoyancy and (iii) weight of the block. Take specific weight of water as 9.79 KN/m³.

Paper 2 of 2 — GET 205 Fundamentals of Fluid Mechanics

Level 200 • First Semester • Handout

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 6

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
- Concepts of velocity distribution, pressure drop, and shear stress in pipe flows
- Fundamentals of dimensional analysis; analysis of selected dimensionless parameters and their significance (Reynolds number as an example)
- Introduction to flow measurement – concepts and measuring devices

2

Fluid Kinematics

➤ Continuity Equation

✓ It's one of the fundamental conservation laws in thermo-fluid studies, highlighting the conservation of mass in thermodynamic systems.

✓ The conservation of mass (continuity) equation defines that mass flow in a system (say a pipe) is the same at the different sections, per adventure there's no gain or loss of matter in the system.

Consider a pipe shown with different cross sections, conservation of mass (continuity)

□□

theorem connotes that $\rho_1 A_1 V_1 = \rho_2 A_2 V_2 = \rho_3 A_3 V_3$, where ρ is density in kg/m^3 , A is

□

□

the cross-sectional area in m^2 , and V is the velocity in m/s .

3

Fluid Kinematics

> Continuity Equation

Therefore, conservation of mass (continuity) theorem connotes that $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$, for flows of both compressible and incompressible fluids.

Recall that density doesn't vary in incompressible fluids, meaning that $\rho_1 = \rho_2$, therefore, the continuity equation reduces to $A_1 V_1 = A_2 V_2$.

Note that the quantity $A_1 V_1$ refers to Rate of flow (or Discharge), or simply volume flow rate of fluid at any section of a pipe/system (m^3/s). It is often denoted by Q , such that $Q = A_1 V_1$.

4

Fluid Kinematics

> Continuity Equation (Differential Form in Cartesian Coordinates)

The conservation of mass (continuity) theorem can also be expressed in differential forms for the different co-ordinate systems.

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

In the cartesian (x, y, z) coordinate system, it can be proven that $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$,

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

where u, v, and w are the velocities of flow in the x, y, and z directions (axes), respectively!

Please review the proof of how the continuity equation is derived in differential forms, for the different co-ordinate systems, just to expand your knowledge of the subject!

5

Fluid Kinematics

> Examples/CPQs

1. For the pipe shown hereunder, solve for the discharge through the pipe and the velocity of water flowing through the pipe at section 1-1, if the velocity at section 2-2 is 1.8 m/s. The diameter of the pipe is 3 m at section 1-1 and 5 m at section 2-2.
2. Consider a 450 mm pipe (1) branching into two pipes (2 and 3) as shown below. Given that 3 m/s of water flows in pipe 1, and 2.8 m/s flows in pipe 2 having 250 mm diameter, solve for the discharge through pipe 1 and the velocity of flow in pipe 3 having 150 mm diameter.

6

Fluid Kinematics

> Examples/CPQs

3. Given that velocities along x and y directions in a 3-dimensional incompressible flow are $u = 2x + 3y$ and $v = -(x^2 + y^2 + z^2)$, determine the velocity along the z direction to satisfy the continuity equation.

4. In a three-dimensional incompressible flow, the velocity components in y and z directions are $v = ax^3 - by^2 + cz^2$ and $w = bx^3 - cy^2 + az^2x$. Determine the third component of velocity distribution such that continuity equation is satisfied.

The concepts of impulse-momentum and energy balance equations apply under fluid dynamics, since both forces and energy driving fluid flow are considered. This should be addressed in the next section of MEE 215!

7

THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lectures 3 - 4

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
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- Introduction to flow measurement – concepts and measuring devices

2

Fluid Statics

➤ Hydrostatic Forces

✓ Fluid statics is used also to determine the forces acting on floating or submerged bodies and the forces developed by devices like hydraulic presses and car jacks, known generally as hydrostatic forces.

✓ The complete description of the resultant hydrostatic force acting on a

submerged surface requires the determination of the magnitude, the direction, and the line of action of the force.

✓ Again, application areas include in the determination of forces acting on surfaces in engineering designs involving stationary fluid, such as in dams and fluid storage vessels.

3

Fluid statics

➤ Hydrostatic forces on submerged surfaces

- On a surface submerged in fluid, hydrostatic forces are developed as the static fluid acts on the surface, whether plane or curved, and we often need to determine the magnitude of the force, known as the total pressure, and its point of application, which is called the center of pressure.

- In most cases, the other side of the plate is open to the atmosphere (such as the dry side of a gate), and thus atmospheric pressure acts on both sides

When analyzing hydrostatic forces

of the plate, yielding a zero resultant. on submerged surfaces, the atmospheric pressure can be

- In such cases, it is convenient to subtract subtracted for simplicity when it atmospheric pressure and work with the gauge acts on both sides of the structure. pressure only.

4

Fluid statics

➤ Hydrostatic forces on horizontally submerged plane surfaces

- Take A = Area of the immersed surface,

x = Depth of horizontal surface from the liquid,

and w = Specific weight of the liquid.

- The total pressure on the surface,

P = Weight of the liquid above the immersed surface

= Specific weight of liquid $\times$ volume of liquid

= Specific weight of liquid $\times$ area of surface $\times$ depth of liquid

= wAx

A plane surface immersed horizontally in a liquid

5

Fluid statics

➤ Hydrostatic forces on vertically submerged surfaces

- Consider a plane vertical surface of arbitrary shape immersed in a liquid, as shown.
- Let, A = total area of the surface,
- G = centre of the area of the surface (centre of gravity),
- h = depth of centre of area,
- h_p = distance of centre of pressure from the free surface of liquid.
- Taking a thin horizontal strip of the surface with width dx and breadth b , it can be shown that the total pressure acting on the surface, $P = wAx$, holds true for all surfaces immersed in liquid, whether plane or curved.

A surface immersed vertically in a liquid

- Also, $P = \rho A h_p$, where ρ = intensity of pressure on the centre of area of the surface.

6

Fluid statics

> Determination of Centre of Pressure

- Since pressure tends to increase with depth in a stationary fluid, the total pressure tends to act at a point on a submerged surface below its centre of gravity. This point of action of the resultant forces of the stationary fluid on the submerged surface, the centre of pressure, is usually expressed as a height from the free liquid surface.
- Please kindly review the mathematical proofs on the determination of the centre of pressure, it is defined by

$$h_p = h + \frac{I_G}{Ah}$$

where I_G is the moment of inertia of the

area

about the

centre of gravity

immersed surface about the free liquid surface, otherwise known as the second moment of area, and h is the moment of inertia of the surface about its centre of gravity.

CPQ - How does h_p relate with h , mathematically?

7

Fluid statics

> Centre of Gravity (G) and moment of inertia (I) of common geometries

Rectangle

Triangle

Circle Trapezium

8

Fluid statics

> Centre of Gravity (G) and moment of inertia (I) of common geometries

9

Fluid statics

> Inclined Immersed Surface

□□

□

• Total pressure, $P = \rho g h_c A$. $\rho g h_c A = \rho g h_c A$

□

□

$\square = \rho g h_c A + \rho g h_c A$

• Centre of pressure, $\square \square$

□□□

An inclined surface immersed in a liquid

10

Fluid statics

> BUOYANCY AND STABILITY

• It is a common experience that an object feels lighter and weighs less in a liquid than it does in air. These and other observations suggest that a fluid exerts an upward force on a body immersed in it. This force that tends to lift the body is called the buoyant force.

• The buoyant force acting on a body of uniform density immersed in a fluid is equal to the weight of the fluid displaced by the body, and it acts upward through the centroid of the displaced volume. This is known as Archimedes' principle.

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Fluid statics

> Stability of Immersed and Floating Bodies

• An important application of the buoyancy concept is the assessment of the stability of immersed and floating bodies with no external attachments.

• We use the classic "ball on the floor" analogy to explain the fundamental concepts of stability and instability. As

shown, three balls at rest on the floor.

- Case (a) is stable since any small disturbance (someone moves the ball to the right or left) generates a restoring force (due to gravity) that returns it to its initial position.
- Case (b) is neutrally stable because if someone moves the ball to the right or left, it would stay put at its new location. It has no tendency to move back to its original location, nor does it continue to move away. 12

Fluid statics

> Stability of Immersed and Floating Bodies

- Case (c) is a situation in which the ball may be at rest at the moment, but any disturbance, even an infinitesimal one, causes the ball to roll off the hill—it does not return to its original position; rather it diverges from it. This situation is unstable.
- What about a case where the ball is on an inclined floor? It is not appropriate to discuss stability for this case since the ball is not in a state of equilibrium.
- In other words, it cannot be at rest and would roll down the hill even without any disturbance.
- For an immersed or floating body in static equilibrium, the weight and the buoyant force acting on the body balance each other, and such bodies are inherently stable in the vertical direction.

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Fluid Statics

> Examples/CPQs

1. Consider a rectangular plate 5 metres long and 2 metres wide immersed vertically in water in such a way that its 5 m side lies in parallel to the water surface and is 1 m below it. Solve for (i) the total pressure on the plate and (ii) the point of action of the total pressure.

14

Fluid Statics

> Examples/CPQs

2. A circular 1m diameter perspex window is on the vertical side of a fish-holding tank containing sea water (see figure below). The top of the window is 2m below the water surface. Determine the magnitude of the total resultant force acting on the window, and the location of the centre of pressure. The moment of inertia of a circle, diameter d , about its centre of gravity axis is $ICG = \pi d^4/64$. For sea water $w = 9.95\text{kN/m}^3$.

15

Fluid Statics

> Examples/CPQs

3. Consider a metallic body floating at the mercury-water interface, shown below, such

that 25 percent of its volume is submerged in mercury, and the remaining in water. Solve for the weight density of the metallic body.

16

Fluid Statics

> Examples/CPQs

4. A boy of weight 400N is floating on a log of wood of weight 600N in sea water. What are the weight and volume of sea water displaced? Take the density of the sea water as 1100kg/m³.

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THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 1

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
- Concepts of velocity distribution, pressure drop, and shear stress in pipe flows
- Fundamentals of dimensional analysis; analysis of selected dimensionless parameters and their significance (Reynolds number, etc)
- Introduction to flow measurement – concepts and measuring devices

2

Learning Objectives (Part 1)

- > To define and characterize fluid/flows and their properties, and fluid mechanics.
- > To differentiate amongst fluid mechanics, fluid kinematics, and fluid dynamics, and apply classical models on each of them to solve simple engineering problems.

➤ To identify the importance of control system/volume definition in thermo-fluid problem-solving and define the same for solving specific simple problems.

➤ To state the classical continuity, momentum, and energy equations (models) and apply the same to solve simple engineering problems.

N.B – Course Learning Outcomes (CLOs) are tied to the learning objectives, with students expected to be able to define and characterize flows and minimum of 5 properties (density, specific volume, specific weight, viscosity, specific heat capacity, etc) in objective 1 above – similar CLOs apply for all the other learning objectives.

3

Course Organization

A 2 credit unit course (2 hours of lectures/recitations per week for 15 weeks).

CA (30 marks) will comprise CPQs/quizzes and problem sets (assignments). Final exam (70 marks). Letter grades as defined in the college handbook.

Flipped classroom and peer instruction teaching/learning approach would be adopted. CPQs could be at the beginning of a class, to assess your participation in the out-of-class activities.

Communications to be handled online via piazza.

Some good textbooks consulted while preparing this and subsequent lecture slides (intellectual credit to authors of these texts):

✓ Fluid Mechanics: Fundamentals and Applications, 4th edition by Yunus A. Çengel and John M. Cimbala.

✓ A Textbook of Fluid Mechanics and Hydraulic Machines by R. K. Rajput

4

Course Outline

✓ Basic Concepts

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- Introduction to flow measurement – concepts and measuring devices

5

General Introduction

How does a wind power plant work? What basic components

and thermo-fluid processes are involved?

6

Introduction - Definitions

> What is fluid mechanics?

What is a fluid? – A substance that flows.

What is mechanics? – A branch of science dealing with the study of objects/processes under the influence of force. Mechanics could be categorized as static (concerning stationary body) or dynamic (relating to body in motion).

Then, what is fluid mechanics?

Fluid mechanics is the science that deals with the behavior of fluid at rest (fluid statics) or in motion (fluid dynamics), and the interaction of fluid with solids or other fluid at the boundaries.

7

Introduction - Definitions

> What is fluid mechanics?

Distinction between a solid and a fluid is made on the basis of the substance's ability to resist an applied shear (or tangential) stress that tends to change its shape.

Stress is defined as force per unit area and is determined by dividing the force by the area upon which it acts.

The normal component of a force acting on a surface per unit area is called the normal stress, and the tangential component of a force acting on a surface per unit area is called shear stress.

In a fluid at rest, the normal stress is called pressure. A fluid at rest is at a state of zero shear stress.

8

Properties of Fluid

> Density

Mass density (or specific mass) of a liquid is defined as mass per unit volume at standard temperature and pressure.

Weight density (or specific weight) is defined as weight per unit volume at standard temperature and pressure, denoted by w , equal to g (9.81 for water).

Specific volume (v) is defined as volume per unit mass at standard temperature and pressure.

What's the relationship between density and specific volume?

9

Properties of Fluid

> Specific gravity

Specific gravity of a fluid is defined as the ratio of its specific weight to that of a standard reference fluid. For liquid, the standard reference fluid is water at 4 degrees C.

> Viscosity

It may be defined as the property of a fluid which determines its resistance to shearing stresses (resistance to flows). It's responsible for the no-slip condition.

What is no-slip condition?

> Example

Calculate the specific weight, density, specific volume, and specific gravity of a liquid having a volume of 8 m³ and weight of 48 kN.

10

Classification of fluid flows

> Classification based on the mode of initiation of fluid motion

Natural (free) flow, when fluid motion is Forced flow, when fluid motion is due to natural means such as the compelled to flow over a surface or in a buoyancy effect; which manifests itself as pipe by external means such as a pump or the rise and fall of warmer (lighter) and a fan. colder (denser) fluids.

> Classification based on variability of flow properties with time

Steady flow, where fluid property at a Unsteady flow, where fluid property at a given location does not change with time. location changes with time.

It's different from uniform flow, where a Turbines, heat exchangers, condensers given flow property is the same at all and pumps work for a long time under the locations at a given time. same conditions, and are steady-flow 11

Classification of fluid flows

> Classification based on the nature of boundary of flow

Internal flow, when fluid flows in a pipe or External flow, when unbounded fluid flows duct completely bounded by solid over a surface such as a plate, a wire or a surfaces. E.g. Flow of crude oil in a pipe. pipe. E.g. Airflow over an exposed pipe Flow of a liquid in duct that is partly-filled during a windy day. is called open-channel flow; such as flow of water in a river.

N.B. Internal flows are dominated by the influence of viscosity throughout the flow field. In external flows, viscous effects are limited to boundary layers near solid surfaces and to wake regions downstreams of bodies.

12

Classification of fluid flows

> Classification based on the level of variation of density during flow

Compressible flow, the case that density of fluid changes during flow. For the same density of fluid remains almost constant quantity of fluid, the volume changes throughout the flow. The volume of every considerably at different sections of flow. portion of fluid remains nearly the same. It applies mostly to gases, although not over the course of its motion. Flows of all flows of all gases can be considered almost all liquids can be considered compressible. incompressible to a very high degree.

Flows of some gases can also be idealized as incompressible to a reasonable level of accuracy, if mach

number, Ma

$Ma < 0.3$.

How would you classify the flow of air at 95 m/s on the basis of density variability, considering that the speed of sound in air at room temperature is 346 m/s? Why? 13

Classification of fluid flows

> Classification based on orderliness of fluid molecules during flow

Laminar flow, where the fluid motion is highly ordered, characterized by smooth layers of fluid. It is common with flow of by high velocities and velocity high-viscosity fluids such as oils at low fluctuations. Example is low-viscosity fluids velocities. such as air at high velocities.

A flow that alternates between being laminar and turbulent is called transitional flow.

14

Classification of fluid flows

> Classification based on the degree of internal resistance to flow

Viscous flow region, where effects of 'internal stickiness' of fluid are significant. are infinitesimally small compared to the inertia of pressure forces and can be some degree. neglected for ease of analysis. It holds true in many flows of practical interest, typically in regions very far from the solid surface.

The flow of an originally uniform fluid stream over a flat plate, and regions of viscous and inviscid flows 15

Classification of fluid flows

➤ Classification based on the number of axis (dimension) of velocity variation

One-dimensional, Two-dimensional, Three-dimensional,
when flow velocity varies along one primary axis.
when flow velocity varies along two primary axes.
when flow velocity varies along the three primary axes. $V(x,y,z)$

Velocity gradient in rectangular
along two dimensions is negligibly small. $V(r,\theta,z)$ in cylindrical
coordinate and negligibly small.

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THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 2

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
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- Introduction to flow measurement – concepts and measuring devices

2

Fluid Properties

➤ Newton's Law of Viscosity

The shear stress (τ) acting on a fluid element layer is directly proportional to the

velocity gradient. The constant of proportionality is called the co-efficient of viscosity or dynamic viscosity (μ). If divided by density, kinematic viscosity (ν) is obtained.

□□

It states mathematically that , □□ being the rate of shear strain, or the velocity gradient induced by the action of the shear stress on the fluid element.

Fluids following the Newton's law of viscosity are known as Newtonian fluids, such as water, kerosene, and air, while those not following the law are generally categorized as non-Newtonian fluids, such as blood, mud flows, polymer solutions, slurries, etc.

Several other types of fluids classify as non-Newtonian, such as ideal fluids, plastic fluids (real plastics and ideal/Bingham plastics), dilatent fluids, thixotropic substances. The following graph defines the stress-velocity gradient relationship for the different types of fluids.

3

Fluid Properties

> Newton's Law of Viscosity

The mathematical expressions for some of the fluid types are:

In case of non-Newtonian fluids, if n is less than unity they are called pseudo-plastics (e.g., paper pulp, rubber suspension paints) while fluids in which n is greater than unity are known as dilatents. (e.g., Butter, printing ink).

Exercise – mention at least 3 examples of other types of non-Newtonian fluids. 4

Fluid Properties

> Newton's Law of Viscosity - CPQs

1. Consider a flat metal strip placed in parallel at the top of a fixed plate at a distance of about 0.000075 m, with hydraulic oil ($\rho = 869 \text{ kg/m}^3$) in between them. Given that the top plate is sheared and kept in motion at a velocity of 1.5 m/s by a force 2.5 N/m^2 . Solve for the dynamic and kinematic viscosities of the hydraulic oil.

2. Given that a flow over a plate is characterized by the velocity distribution, $u = u_{\max} - \frac{y^2}{2\delta^2} u_{\max}$, where u is the velocity in m/s of the flow at a distance y m above the plate. Determine the flow velocity gradient and the shear stress right on the surface of the plate and at a height 1.8 m above it. Assume $0.9 \text{ Pa}\cdot\text{s}$ for the dynamic viscosity of the fluid.

5

Fluid Statics, Kinematics, and Dynamics

> Overview of Fluid Statics, Kinematics, and Dynamics

✓ Fluid Statics

- It deals with problems associated with fluids at rest. There is no relative motion between adjacent fluid layers, and thus there are no shear (tangential) stresses in the fluid trying to deform it.
- The only stress in consideration in fluid statics is the normal stress, which is the pressure, and the variation of pressure is due only to the weight of the fluid.
- Thus, the topic of fluid statics has significance only in gravity fields, and it concerns principally the study of pressure (normal forces) and its effects on engineering bodies.
- The design of many engineering systems such as water dams and liquid storage tanks requires the determination of the forces acting on their surfaces using fluid statics.
- Fluid statics is generally referred to as hydrostatics when the fluid is a liquid and as aerostatics when the fluid is a gas.

6

Fluid Statics, Kinematics, and Dynamics

> Overview of Fluid Statics, Kinematics, and Dynamics

✓ Fluid Kinematics

- It deals with the study of how fluid flows, that is, the velocity and acceleration of particles of fluid in motion and their distribution in space without considering any force or energy involved.
- The motion of fluid particles can be described by using either the Lagrangian method or the Eulerian method.

✓ Fluid Dynamics

- Like fluid kinematics, fluid dynamics also deals with the study of flow of liquids and gases. Thus, fluid kinematics can be said to be a branch of fluid dynamics.
- However, unlike in fluid kinematics where forces and energy causing the flow are not considered, fluid dynamics studies the flow of fluids under the influence of forces and energy.

More attention is given here to fluid statics as described in the following slides, but very basic concepts of fluid kinematics and dynamics are also defined herein and in later topics under conservation laws and equations. 7

Fluid Statics

> Pressure

✓ Definition

- When a fluid is contained in a vessel, it exerts force at all points on the sides and bottom and top of the container. The force per unit area is called pressure.

□

- If, F = the force, and A = Area on which the force acts; then pressure, $\square =$

□

- Again, the pressure of a fluid on a surface will always act normal to the surface.
- In a liquid, pressure builds up due to the weight of the fluid, and this pressure increases as the depth of the liquid increases.

For a liquid-filled container shown, with cylinder inserted, the total pressure at the bottom of the pressure would correlate with the height of the cylinder, equal to the weight of the liquid in the cylinder. Mathematically, $P = \rho h = \rho g h$.

8

Fluid Statics

> Governing Principles/Laws

✓ Hydrostatic law

- The rate of increase of pressure in a vertically downward direction must be equal to the specific weight of the fluid at that point, $\frac{dP}{dh} = \rho g$.

✓ Pascal law

- The intensity of pressure at any point in a liquid at rest, is the same in all directions.

> Pressure Forms / Indications

✓ Atmospheric pressure

- It's the normal pressure exerted by the atmospheric air upon all surfaces with which it is in contact, otherwise known as Barometric pressure.
- The atmospheric pressure at sea level (above absolute zero) is called 'Standard atmospheric pressure.'

9

Fluid Statics

> Pressure Forms / Indications

✓ Gauge pressure

- It is the pressure, measured with the help of pressure measuring instrument, in which the atmospheric pressure is taken as datum. The atmospheric pressure on the scale is marked as zero.
- Gauges record pressure above or below the local atmospheric pressure, since they measure the difference in pressure of the liquid to which they are connected and that of surrounding air.
- If the pressure of the liquid is below the local atmospheric pressure, then the gauge is designated as 'vacuum gauge' and the recorded value indicates the amount by which the pressure of the liquid is below local atmospheric pressure, i.e. negative pressure, otherwise known as vacuum pressure.

10

Fluid Statics

> Pressure Forms / Indications

✓ Absolute pressure

- It is necessary to establish an absolute pressure scale which is independent of the changes in atmospheric pressure. A pressure of absolute zero can exist only in complete vacuum.
- Any pressure measured above the absolute zero of pressure is termed 'absolute pressure.'

$$P_{abs} = P_{atm} + P_{gauge}$$

$$P_{gauge} = P_{abs} - P_{atm}$$

Pressure is measured fundamentally in Newton per square metre (N/m²), known also as Pascal. Low pressures are often expressed in terms of mm of water or mm of mercury. This is an abbreviated way of saying that the pressure is such that will support a liquid column of stated height. Relationship between pressures

11

Fluid Statics

> Pressure Measurement – achieved using manometers or mechanical gauges

✓ Manometers

- Devices used for measuring the pressure at a point in a fluid by balancing the column of fluid by the same or another column of liquid. These are classified as (a) Simple manometers, such as (i) Piezometer, (ii) U-tube manometer, (iii) Single column manometer, and (b) Differential manometers for measuring pressure difference between two points in a pipe or two pipes, such as (i) U-tube differential manometer, (ii) inverted U-tube differential manometers.

Vertical single column U-tube differential manometer

Piezometer U-tube manometer manometer 12

Fluid Statics

> Pressure Measurement – achieved using manometers or mechanical gauges

✓ Mechanical Gauges

- Devices in which the pressure is measured by balancing the fluid column by spring (elastic element) or dead weight. Generally these gauges are used for measuring high pressure and where high precision is not required. Examples are Bordon tube pressure gauges and Diaphragm gauges.

Bordon tube pressure gauge Diaphragm gauge 13

Fluid Statics

> Examples/CPQs

1. A U-tube manometer is used to measure the pressure of oil of specific weight 8.34 kN/m^3 flowing in a pipe line. Its left end is connected to the pipe and the right limb is

open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific weight = 133.42 kN/m^3) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.

14

Fluid Statics

> Examples/CPQs

2. A U-tube differential manometer connecting two pressure pipes at A and B is shown below. The pipe A contains a liquid of specific gravity 1.8 under a pressure of 130 kN/m^2 . The pipe B contains oil of specific gravity 0.7 under a pressure of 220 kN/m^2 . Find the difference of pressure measured by mercury as fluid filling U-tube.

15

Fluid Statics

> Examples/CPQs

3. A U-tube differential manometer connected at two points A and B is shown below. Given that air pressure at point A is 140 kN/m^2 , solve for the absolute pressure at point B.

16

Fluid Statics

> Examples/CPQs

4. An inverted differential manometer is shown below, containing an oil of specific gravity 0.8 connected to two different pipes carrying water under pressure. Given that the pressure in the pipe A is 20 kN/m^2 , determine the pressure in pipe B.

17

THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 5

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows

- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications
- ✓ Characterisation and Measurement of Flows
- Introduction to incompressible viscous flow of Newtonian fluids in pipes
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- Introduction to flow measurement – concepts and measuring devices

2

Fluid Kinematics

> Fluid Kinematics

- ✓ Again, fluid kinematics concerns the study of fluid flow and description of fluid motion without any recourse to the forces/energy driving the flow.
- ✓ Lagrangian and Eulerian methods are the two prominent approaches commonly used to describe fluid motion in fluid kinematics.
- ✓ Description of fluid motion using the Lagrangian method requires that the position (path taken), velocity and acceleration of each fluid particle be tracked within the flow. Invariably, a mass of fixed identity is followed within the flow, analogous to what obtains in a closed (or control mass) thermodynamic system.

CPQs – What's a thermodynamic system? When is a system said to be closed, open, or isolated?

3

Fluid Kinematics

> Fluid Kinematics

- ✓ In the Eulerian method, a finite volume, known generally as control volume, is first defined through which fluid flows in and out. The control volume is analogous to the open thermodynamic system, illustrated hereunder.
- ✓ Then, instead of tracking fluid particles as done in the Lagrangian method, field (point) variables are defined within the control volume as functions of space and time.
- ✓ The field variable at a particular location at a particular time is the value of the variable for whichever fluid particle happens to occupy that location at that time. Examples are velocity and acceleration fields, defined as vectors below.

A typical control volume

4

Fluid Kinematics

➤ Flow Lines

✓ Fluid particles in motion tend to follow some defined lines, known generally as flow lines.

✓ Prominent among the different types of flow lines are: i) Path Line, ii) Stream Line, iii) Stream Tube, and iv) Streak Line.

✓ Please compare and contrast between the different types of flow lines listed above, with the aid of diagrams.

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Fluid Kinematics

➤ Examples/CPQs

1. Solve for the velocity and acceleration at a point (2, 4, 6) after 2 seconds in a 3-dimensional flow given by $u = yz + t$, $v = xz - t$, $w = xy$ m/s.

2. Given the velocity field in fluid flow as $V = (6x+4y)i + (4z+6x^2)j + (t-2k)z$. Determine: i) the velocity component u, v and w at any point in the flow field, (ii) the speed at point (1,1,1), and (iii) the speed at time $t = 2s$ at point (0,0,4). Would you say the velocity field is steady or unsteady, uniform or non-uniform, 1-dimensional, 2-dimensional, or 3-dimensional?

Justify your answer!

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THANK YOU FOR YOUR ATTENTION!

Any other questions?